MLOPS ASSIGNMENT-1

Apache Pig

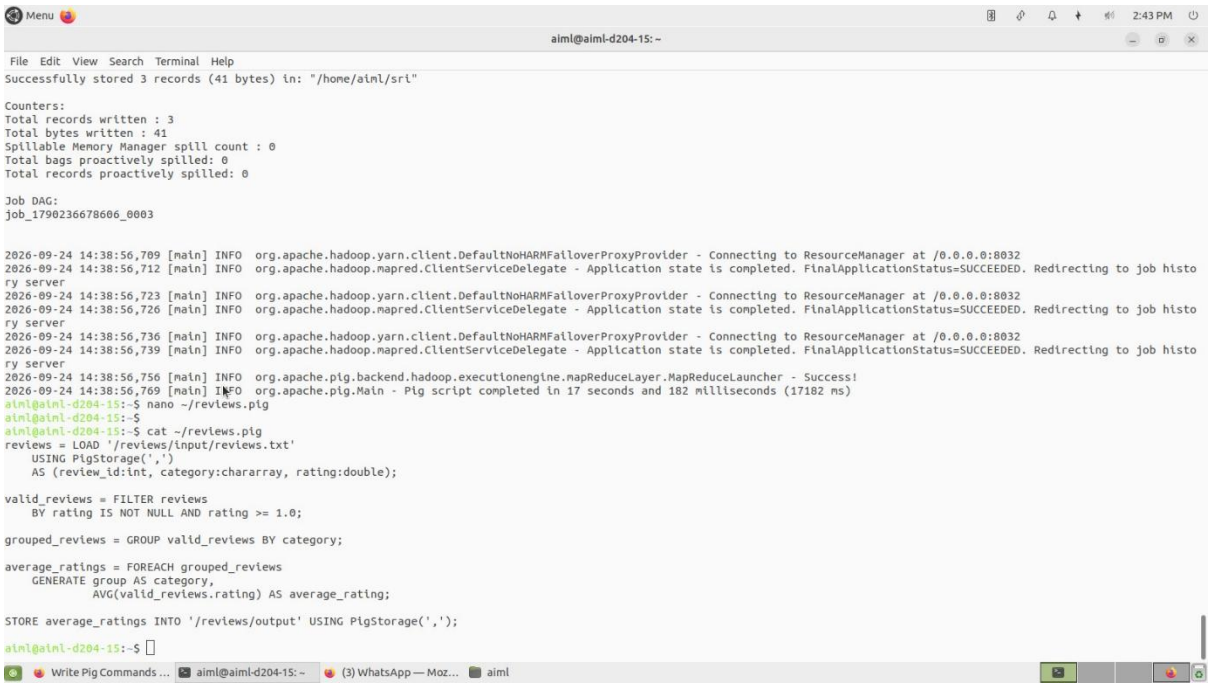
PROBLEM STATEMENT

An e-commerce company has a dataset of product reviews containing review ID, product category, and rating. Some reviews contain invalid ratings below 1.0 or missing ratings. The objective is to use Apache Pig and Hadoop to filter out these invalid reviews, group the valid reviews by product category and calculate the average rating for each category.

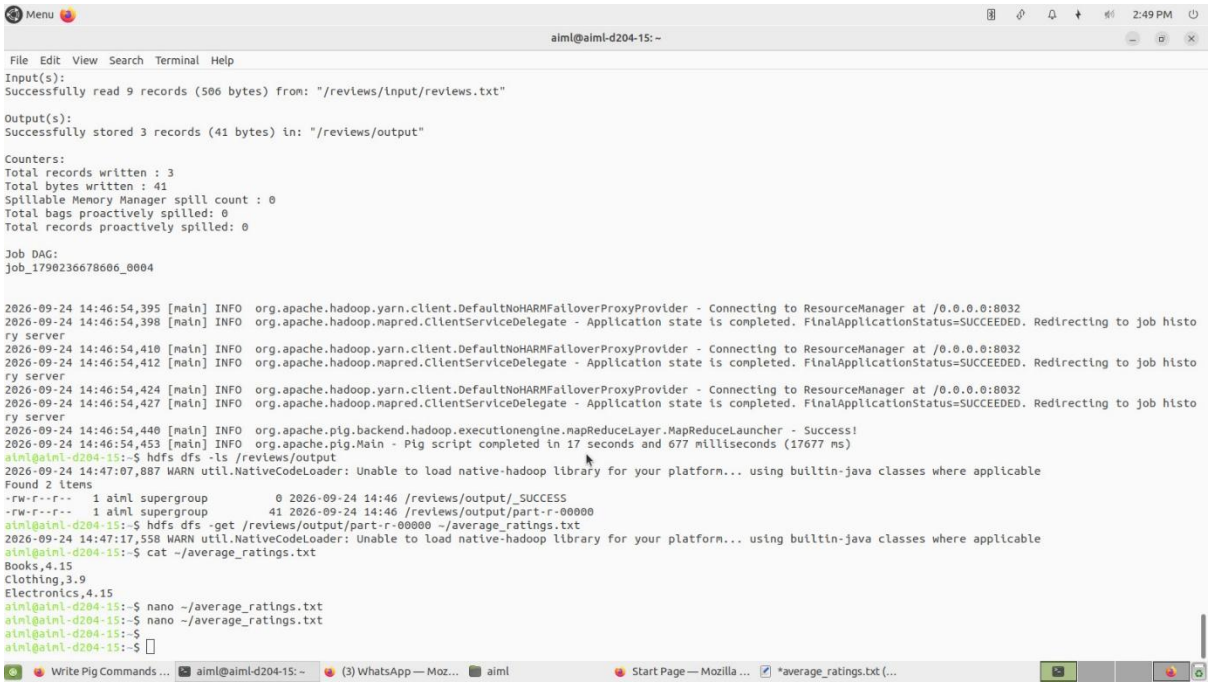
CODE IMPLEMENTATION

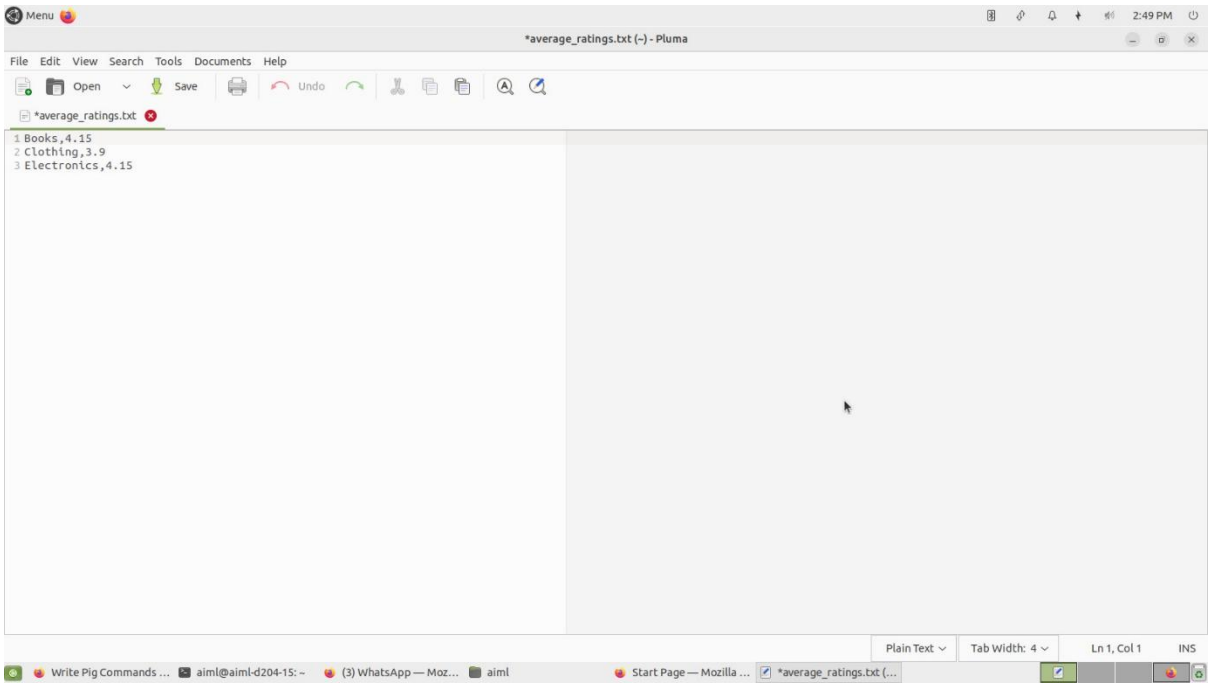


reviews.pig



OUTPUT





Steps:

1. Create the review dataset with review ID, category, and rating.
2. Store the dataset in HDFS using Hadoop commands.
3. Load the data in Pig using LOAD and PigStorage(',').

4. Filter invalid reviews where the rating is below 1.0 or null.
5. Group valid reviews based on product category.
6. Calculate average rating for each category using AVG().
7. Run the Pig script using Hadoop MapReduce through YARN.

TOOLS

- Ubuntu Linux
- Java 8
- Apache Hadoop 3.3.6
- HDFS
- YARN / MapReduce
- Apache Pig 0.17.0
- Pig Latin

Result

The Apache Pig program successfully filtered invalid ratings and calculated the average rating for each product category.

Category	Average Rating
Books	4.15
Clothing	3.90
Electronics	4.15

Conclusion

The Apache Pig program successfully filtered invalid product reviews and calculated the average rating for each product category. The results were processed using Hadoop and stored in HDFS, achieving the objective of the problem statement.

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